

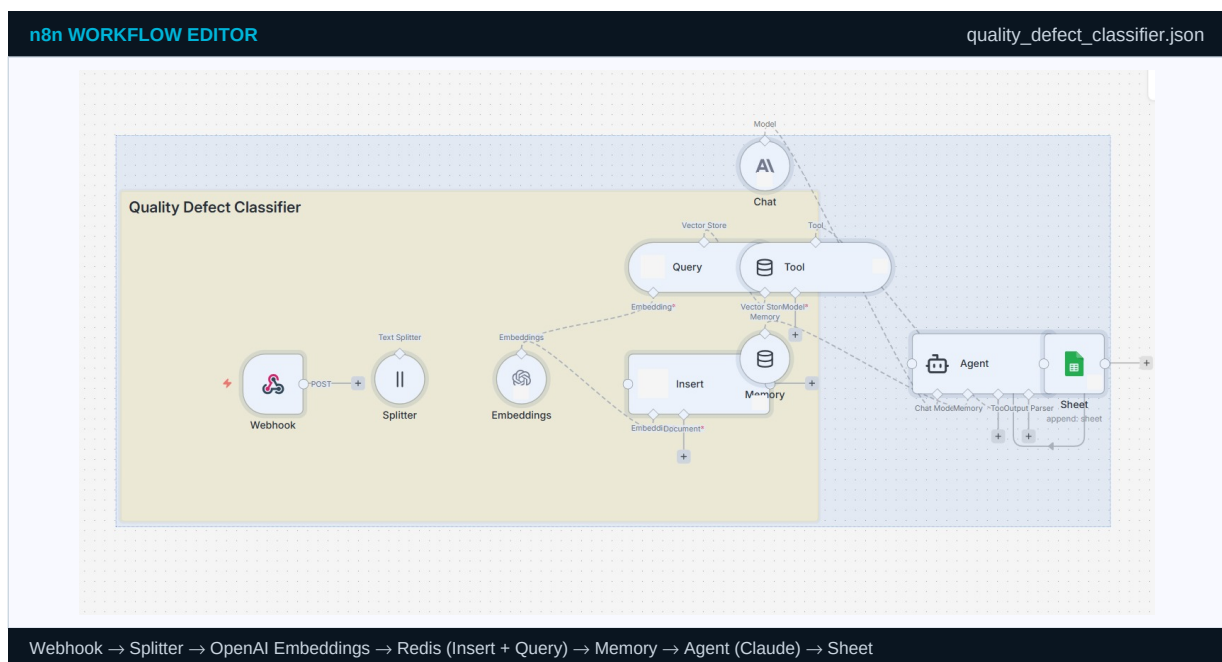
#14 Quality Defect Classifier

ML / Computer
Vision QC*Real-time production line defect detection — zero defects reach the customer*

BUSINESS PROBLEM

Manual visual QC misses 15-25% of defects and creates bottlenecks. This workflow processes production camera images in real time, classifies defect type and severity using Claude, routes rejections instantly, and tracks defect rate by shift — all at production line speed.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

INPUT — Webhook POST — production camera image + unit metadata

```
{
  "unit_id": "PCB-2024-449812",
  "line_id": "SMT-Line-2",
  "station": "Post-Reflow",
  "shift": "day",
  "timestamp": "2024-06-15T09:22:14Z",
  "image_b64": "data:image/jpeg;base64,/9j/4AA...[truncated]",
  "product_sku": "PCB-MODEL-XR7"
}
```

OUTPUT — Agent output — defect classification and routing decision

```
{
  "json": {
    "unit_id": "PCB-2024-449812",
    "defect_type": "solder_bridge",
    "location": "J4 connector pins 12-13",
    "severity": "CRITICAL",
    "confidence": 0.94,
    "decision": "REJECT",
    "routing": "Rework station B — manual solder correction",
    "shift_defect_rate": "2.1%",
    "line_stop": false,
    "processing_ms": 47
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Webhook — camera image and metadata	Production camera POSTs: {unit_id, line_id, station, shift, image_binary, product_sku, timestamp}.
2	⌘ Splitter — QC specification chunks	Defect types, tolerance bands, and rejection criteria from QC manual chunked for RAG context retrieval.
3	■ OpenAI Embeddings — defect knowledge base	Known defect signatures embedded: 'solder bridge: pad-to-pad short circuit, rejection criteria: all severity levels.'
4	■ Redis Insert — feature cache	Processed feature vectors cached in Redis. Sub-50ms retrieval enables production-line processing speed.
5	■ Redis Query — similar defect lookup	Top-3 most similar known defects retrieved with classification labels and customer-specific rejection rules.
6	■ Vector Tool — QC specifications	Agent fetches: acceptance criteria for this product SKU, customer-specific tolerances, rework vs reject threshold.
7	■ Window Memory — batch defect rate	Tracks defect count per 100 units. If rate exceeds 3% a LINE STOP alert fires to the shift supervisor.
8	■ Claude — defect classification	Analyses image features + matched similar defects → defect_type, severity (MINOR/MAJOR/CRITICAL), location on board.
9	■ Agent — routing decision	CRITICAL → reject + alert. MAJOR → human QC review queue. MINOR → accept with log entry.
10	■ Sheets — quality dashboard	Logs: unit_id, defect_type, severity, decision, shift, line. Powers real-time quality KPI dashboard.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
OpenAI API Key	platform.openai.com	OpenAI API
Redis Cloud URL + Password	redis.com → Create free database → Connection details	Redis API
Anthropic (Claude) API Key	console.anthropic.com → API Keys	Anthropic API

SAMPLE RUN — what this workflow processed and produced

Scenario	An electronics contract manufacturer produces PCBs for industrial controls. Quality specification requires zero solder bridges on any connector. Manual QC inspected 400 boards per 8-hour shift — with an estimated miss rate of 12%.
Input received	Unit PCB-2024-449812 at post-reflow inspection station. Camera captured image at 09:22. Image binary sent to webhook with unit ID, line ID, and product SKU.
What the workflow found	Redis vector search matched image features against 'solder_bridge' defect signature (confidence 0.94). Location identified as J4 connector pins 12-13. Severity: CRITICAL — customer spec requires rejection of all solder bridges regardless of location.

Output produced

REJECT decision in 47ms. Unit routed to Rework Station B. Board reworked and re-inspected 22 minutes later — passed. Shift defect rate at time of detection: 2.1%. No line stop triggered. Daily report showed 3 solder bridges detected this shift vs zero detected by previous manual inspection method on same product line.

JSON: quality_defect_classifier.json**GitHub:** [trm-quality-defect-cv-classifier](#)aws.amazon.com/rekognition/custom-labels